DCIT 316 · FINAL INDIVIDUAL RESEARCH PROJECT

Tokenization Efficiency in Ghanaian Low-Resource Languages

Comparative evaluation of modern multilingual tokenizers for Twi and Ewe

Jennifer Banibensu · Student ID 22013023

Department of Computer Science · University of Ghana · UG
Repository: github.com/exornamb/GhanaToken · final tag v1.0-final

Made with GAMMA

THE PROBLEM

Multilingual tokenizers create a token tax for Twi and Ewe

Token fertility = average tokens per whitespace-separated word.

On sealed test sentences, existing tokenizers use **2.29–2.61** tokens per word and split **64%–72%** of words.

More tokens can raise usage cost and reduce how much text fits in a fixed context window.

Ghanaian languages are underserved; this study tests the problem side by side on the same Twi and Ewe data.

Made with GAMMA

EVALUATION FRAME

Three questions define the evaluation

01 RQ1

How do mBERT, XLM-R, AfroXLMR, and ABENA fragment Twi and Ewe?

02 RQ2

Does GhanaTok reduce fertility and fragmentation on sealed test data?

03 RQ3

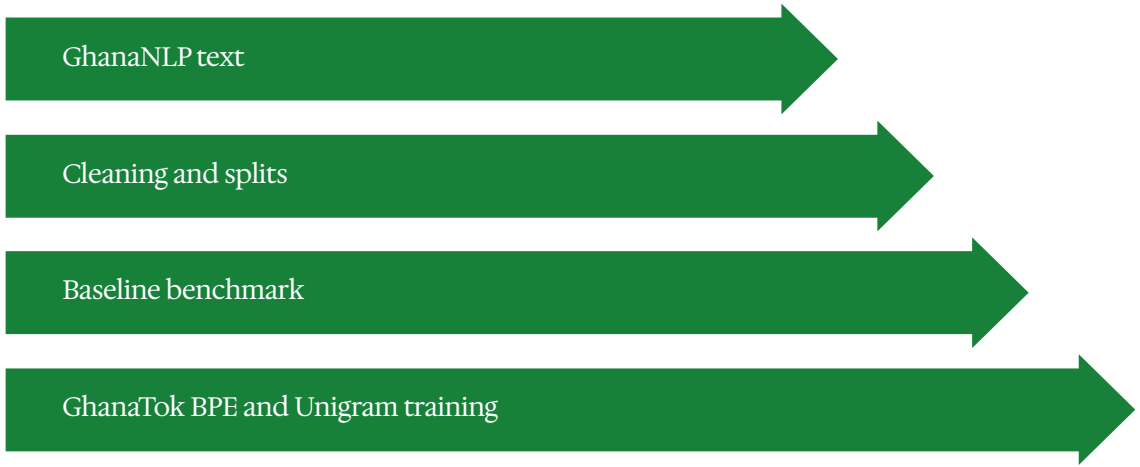
Can one shared 8,000-token vocabulary work effectively for both languages?

Scope boundary. Lower fertility is not treated as proof of better language understanding or downstream accuracy.

Made with GAMMA

METHOD

A controlled pipeline isolates tokenizer efficiency



CORPUS

6,090 raw Twi · **3,488** raw Ewe

After cleaning: 6,072 Twi · 3,456 Ewe

SPLIT

80/10/10

608 Twi · 347 Ewe sealed test sentences

TRAINING

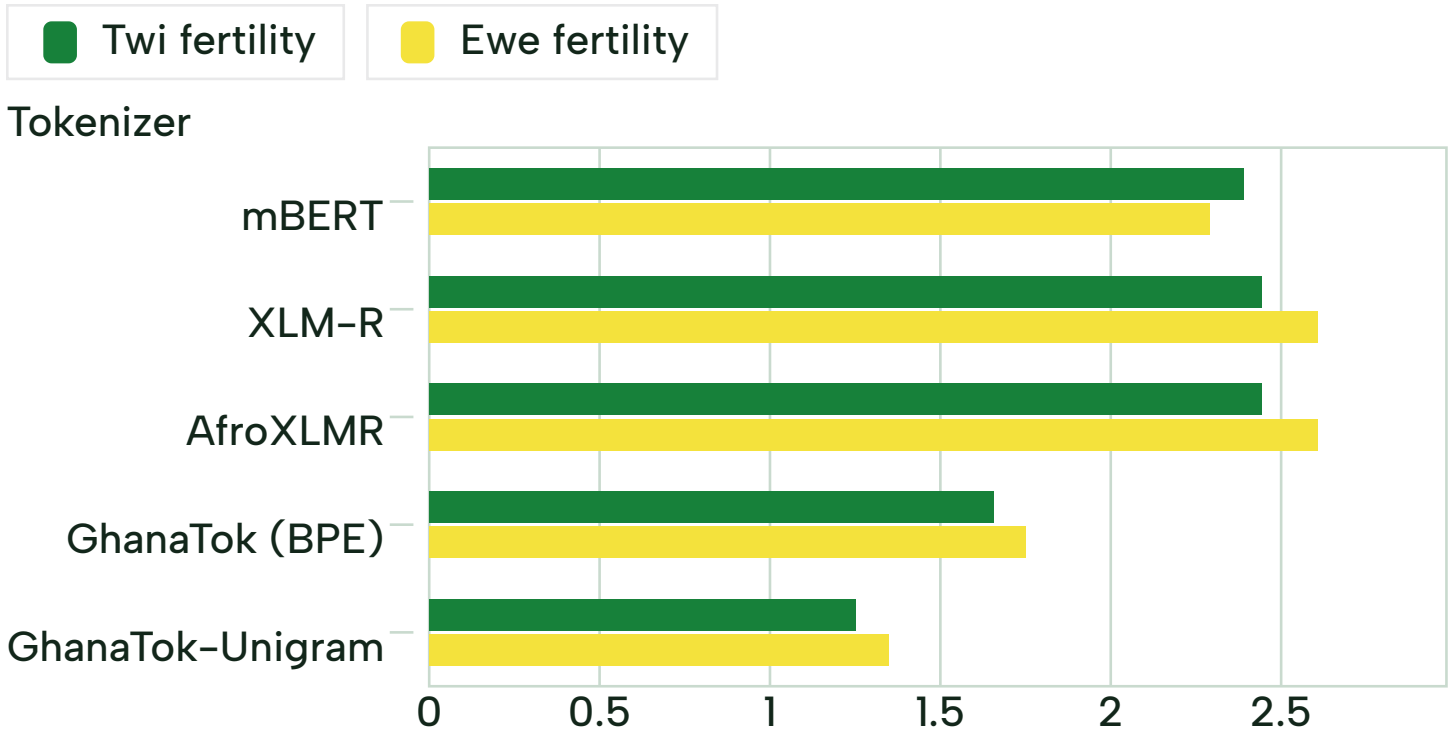
7,621 combined training sentences

Letter-aware handling keeps ɛ, ɔ, ɖ, ŋ, ʊ, *ƒ* and tone marks attached.

Made with GAMMA

SEALED TEST · LOWER IS BETTER

GhanaTok sharply lowers fertility on the sealed test



Specialization cuts fertility.

Against mBERT, BPE reduces fertility by **30.7%** on Twi and **23.6%** on Ewe. Twi-only ABENA: **2.419** on Twi. Unigram reaches **1.251** on Twi and **1.349** on Ewe: **47.6%** and **41.1%** below mBERT.

Shared 8,000-token vocabulary; main version: byte-level BPE; Unigram: secondary comparison.

Made with GAMMA

ROBUSTNESS

The gains are consistent across sentence distributions

PAIRED COMPARISONS

Paired bootstrap intervals use **1,000 resamples**; paired Wilcoxon signed-rank tests compare tokenizers on the same sentences.

MULTIPLE TESTING

$p < 10^{-50}$

All five main GhanaTok BPE comparisons remain significant after Holm correction.

LANGUAGE GAP

0.094 tokens per word

Twi–Ewe BPE gap, with a 95% interval of **0.061–0.128**.

INTERPRETATION

The measured reduction is robust for these test distributions, not a universal claim about every domain or dialect.

Content differences may contribute because test sentences differ.

Made with GAMMA

SENTENCE-LEVEL RESULTS

Specialization wins broadly, but mixed-language text exposes the trade-off

TWI

598 / 608

98.4% of sentences

GhanaTok BPE uses fewer tokens than mBERT; it ties on **5** and loses on **5**.

EWE

330 / 347

95.1% of sentences

GhanaTok BPE wins and loses on **13**.

Where losses cluster

English phrases, loanwords such as *fisiks*, quotation marks, digits, and mixed-language text.

Likely explanation: GhanaTok spends vocabulary capacity on Twi and Ewe; English coverage is thinner. This pattern is consistent with the data, not a proven causal decomposition.

Made with GAMMA

DOWNSTREAM PROBE

Fewer tokens did not separate the sentiment probe

Tokenizer group	Accuracy	Macro F1	Active features
GhanaTok BPE	62.91%	0.536	1,899
GhanaTok-Unigram	62.91%	0.536	2,133
Baselines	64.07%–65.02%	Intervals overlapped	3,886–4,483

HONEST CAVEAT

This was a small bag-of-tokens logistic-regression probe on short tweets, not a language-model quality test.

Token count ≠ model intelligence. Lower fertility can improve efficiency without proving better understanding.

Made with GAMMA

NEXT WORK

The result is promising, not complete

01 Add English and mixed-language coverage without losing Ghanaian-language specialization.

02 Pre-train a small language model with GhanaTok and measure perplexity, training cost, and task accuracy.

03 Test news, books, social media, additional dialects, and larger held-out samples.

04 Run ablations for the letter-aware pre-tokenizer and seed morphemes; add native-speaker review of linguistic plausibility.

05 Small and narrow test sets, same-corpus training advantage, no ablations, one training run, and rough cost/speed estimates.

Made with GAMMA

CONCLUSION

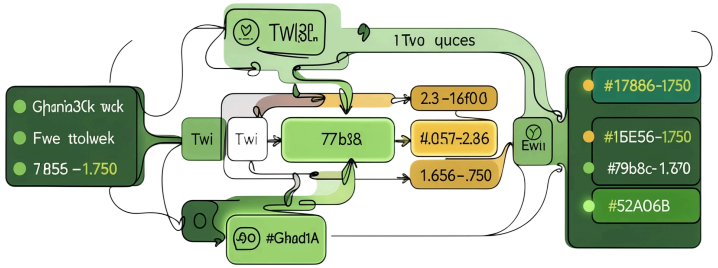
GhanaTok removes a measurable inefficiency

—

Existing multilingual tokenizers fragment most Twi and Ewe words: **2.3–2.6** tokens per word.

A shared **8,000-token** GhanaTok brings fertility to **1.656** on Twi and **1.750** on Ewe; Unigram reaches **1.251** and **1.349**.

The gain holds on nearly every sentence, while failures reveal the cost of narrow language specialization.



DELIVERABLES

Reproducible code, notebooks, saved metrics, figures, documentation, and final tag **v1.0-final**.

Repository: github.com/exornamb/GhanaToken

Made with GAMMA